

Lab Guide 9

Kubernetes Scheduling

Q.1 Create a pod named test1 using httpd image in node3 machine

Q.2 Create a pod named test1 using httpd image in node2 machine (Using Node selectors)

Q.3 Create a pod with toleration values mayank=test:NoSchedule

Ans: 1

```
kubectl run test1 --image httpd --dry-run -o yaml > nodenamepod.yaml
```

```
vim nodenamepod.yaml
```

```
cat nodenamepod.yaml
```

```
kubectl create -f nodenamepod.yaml
```

```
kubectl get pods -o wide
```

```
root@master:~# kubectl run test1 --image httpd --dry-run -o yaml > nodenamepod.yaml
W0525 03:41:41.851820 1777 helpers.go:663] --dry-run is deprecated and can be replaced with --dry-run=client.
root@master:~# vim nodenamepod.yaml
root@master:~# cat nodenamepod.yaml
apiVersion: v1
kind: Pod
metadata:
  creationTimestamp: null
  labels:
    run: test1
  name: test1
spec:
  nodeName: node3
  containers:
  - image: httpd
    name: test1
    resources: {}
  dnsPolicy: ClusterFirst
  restartPolicy: Always
  status: {}

root@master:~# kubectl create -f nodenamepod.yaml
pod/test1 created
root@master:~# kubectl get pods -o wide
NAME    READY   STATUS    RESTARTS   AGE   IP            NODE    NOMINATED NODE   READINESS GATES
test1   1/1     Running   0           3s    192.168.3.33  node3   <none>            <none>
root@master:~#
```

Ans: 2

Label a node with any key=value pair

kubectl label node node2 mayank=test

kubectl run test1 --image httpd --dry-run -o yaml > nodeselepod.yaml

vi nodeselepod.yaml

cat nodeselepod.yaml

kubectl create -f nodeselepod.yaml

kubectl get pods -o wide

```
root@master:~# kubectl label node node2 mayank=test
node/node2 labeled
root@master:~# kubectl run test1 --image httpd --dry-run -o yaml > nodeselepod.yaml
W0525 03:55:35.753355 2240 helpers.go:663] --dry-run is deprecated and can be replaced with --dry-run=client.
root@master:~# vi nodeselepod.yaml

root@master:~# vi nodeselepod.yaml
root@master:~# kubectl create -f nodeselepod.yaml
pod/test1 created
root@master:~# kubectl get pods -o wide
NAME      READY   STATUS    RESTARTS   AGE   IP            NODE    NOMINATED NODE   READINESS GATES
test1     1/1     Running   0           4s    192.168.2.47  node2   <none>            <none>
root@master:~# cat nodeselepod.yaml
apiVersion: v1
kind: Pod
metadata:
  creationTimestamp: null
  labels:
    run: test1
  name: test1
spec:
  nodeSelector:
    mayank: test
  containers:
  - image: httpd
    name: test1
    resources: {}
  dnsPolicy: ClusterFirst
  restartPolicy: Always
status: {}
root@master:~#
```

Ans: 3

kubectl taint node node3 mayank=test:NoSchedule

vim pod.yaml

kubectl create -f pod.yaml

kubectl get pods -o wide

```

root@master:~# cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  creationTimestamp: null
  labels:
    run: mayank
  name: mayank
spec:
  containers:
  - image: httpd
    name: mayank
  tolerations:
  - key: mayank
    value: test
    operator: Equal
    effect: NoSchedule

root@master:~# █

```

```

root@master:~# vim pod.yaml
root@master:~# kubectl get node
NAME                STATUS    ROLES    AGE   VERSION
ip-172-31-13-153    NotReady control-plane 25h   v1.26.1
ip-172-31-6-5       NotReady <none>      25h   v1.26.1
node2               Ready     <none>      25h   v1.26.1
node3               Ready     <none>      25h   v1.26.1
root@master:~# kubectl describe node node3 | grep -i taint
Taints:      mayank=test:NoSchedule
root@master:~# vim pod.yaml
root@master:~# kubectl delete pod mayank
pod "mayank" deleted
^Croot@master:~# kubectl create -f pod.yaml
pod/mayank created
root@master:~# kubectl get pods -o wide
NAME                READY    STATUS    RESTARTS   AGE   IP              NODE    NOMINATED NODE   READINESS GATES
mayank              1/1      Running   0           4s    192.168.3.53    node3    <none>            <none>
mayank-89b68f7f6-nsq7c 1/1      Running   0          2m39s  192.168.2.83    node2    <none>            <none>
root@master:~# cat pod.yaml
root@master:~# cat pod.yaml
apiVersion: v1
kind: Pod

```